

Binning and significance fields

Independent diagnostic report, version 5.1.0

9 September 2026

What this study establishes

The existing signed likelihood-root scan can be the statistic used by a significance GP. The unresolved question is its distribution under a credible background, not whether an observed significance curve is available. This report tests actual histogram rebinning, recalculates the corresponding full response matrices, changes the mass-scan grid, and studies illustrative correlations between input bins. All choices are reported; none is adopted because it increases the observed significance.

The raw excess peaks remain near 51 MeV in 2015 and 78–79 MeV in 2021; the largest 2016 root switches between 90 and 117 MeV. Coarsening the histogram does not remove the large 2016 stress bias. These results identify stability and model dependence, not a more reliable high-significance signal. The follow-up on pages 7–8 tests 32/33 MeV support edges and explains the 2016 offset using interpolation residuals and 64 paired Poisson spectra.

Two different grids, two different correlations

Histogram bins contain candidate counts. Independent events each assigned to one disjoint bin are compatible with independent Poisson counts when the total event count is Poisson. Detector mass smearing alone does not invalidate this model. Multiple candidates from the same event, weighted or overlapping entries, and correlations between events can invalidate it. If the total count is fixed, a multinomial model instead has negative off-diagonal covariances. A shared uncertain background shape is another source of dependence, whose treatment depends on the nuisance model.

Mass hypotheses reuse events and sidebands. Their fitted significances are expected to be correlated even when the input count bins are independent. The GP prediction covariance and the significance-field covariance therefore do not, by themselves, demonstrate correlations between the input counts. The three source histograms have stored variances equal to their counts. This is compatible with unit event weights; it does not verify event-level independence or exclude multiple candidates per event.

Where the paper’s perturbations belong

The paper [1] perturbs an expected background spectrum one input bin at a time, then refits the entire signed scan. Its canonical field has zero null mean and unit variance. Writing ℓ_p for the profiled negative log-likelihood, the HPS stress construction gives

$$r_m = \text{sgn}(\hat{A})\sqrt{2[\ell_p(0) - \ell_p(\hat{A})]}, \quad a_m = r_m(B), \quad D_{im} = r_m(B + \sqrt{B_i}e_i) - a_m.$$

We retain $C = D^T D$, $s_m = \sqrt{C_{mm}}$ and $R_{mn} = C_{mn}/(s_m s_n)$. Centering here is an explicit conditional extension of the paper. It does not establish that the stress spectrum is the correct background. Adding noise directly to the observed curve would center the experiment on the data, including possible signal and modeling residuals.

The weak/strong toy probabilities are about upper-limit strength. They do not replace the discovery statistic. Their precise definitions and the valid alternatives are given on page 6.

1. Actual histogram binning: exact pointwise fits

The baseline bins are 0.25 MeV for 2015/2016 and 0.625 MeV for 2021. Both twice and four times those widths are tested. Counts, sideband masks, signal-bin integrals and GP predictions are recalculated. Reviewed kernel hyperparameters and other settings remain fixed at each mass. This tests binning with a fixed kernel policy; it is not a retuned analysis.

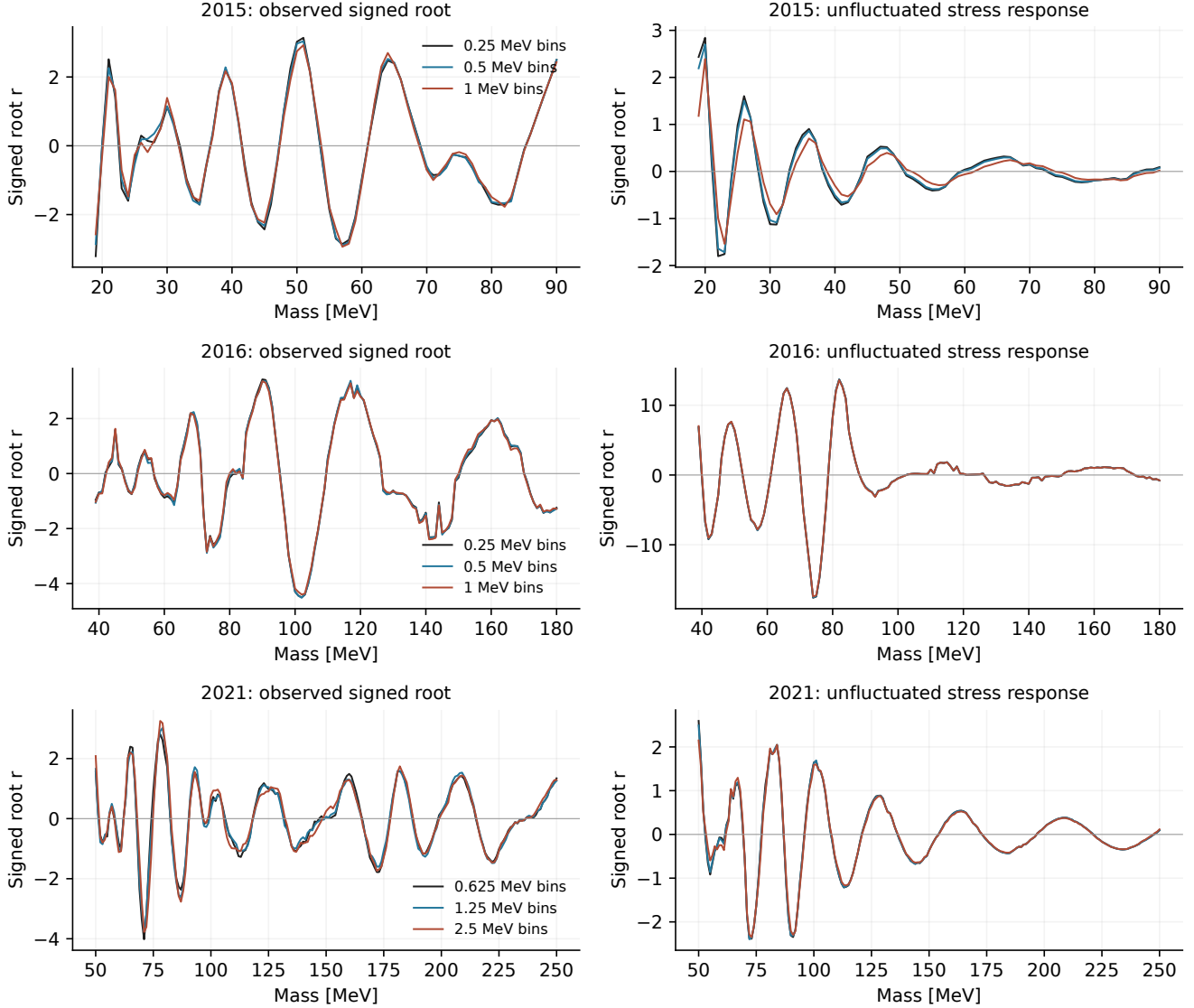


Figure 1. Observed signed roots (left) and unfluctuated stress responses (right). Every original integer mass is retained, including negative roots. The baseline reuses 415 archived values with nine exact spot checks; 830 coarse-bin mass coordinates are newly evaluated for both observed and stress spectra. These are raw profiled re-evaluations associated with the global studies, not changes to the frozen released asymptotic results.

For 2015 the leading root changes from 3.139 to 3.039 and 2.927 at 51 MeV. For 2021 it changes from 2.809 at 78 MeV to 3.017 at 79 MeV and 3.254 at 78 MeV. The 2016 maximum changes from 3.425 at 90 MeV to 3.374 at 117 MeV and 3.357 at 90 MeV. The 2016 stress RMS barely changes: 4.967, 4.959 and 4.933. Choosing the largest value after these comparisons would add a binning-selection trial; no such combined search is calibrated.

2. Fresh significance-GP calculations for the coarser histograms

Both coarse choices receive their own complete one-bin perturbation basis, using the same coherent stress spectrum at every hypothesis. Each perturbed spectrum is refitted with the exact cached GP equations and profiled likelihood. We do not substitute the old correlation matrix for a new histogram binning. Scalar checks and likelihood-score checks accompany every new mass coordinate; the original baseline matrices are retained separately.

For continuity with v5.0.2, the conditional ordering is

$$z_m = (r_m - a_m)/s_m, \quad T = \max_{m:r_m>0} z_m.$$

For each binning, the global probability is evaluated over its entire mass grid. The selected conditional threshold is integrated with 20,000 importance-sampled complete Gaussian fields [3]. This avoids zero-count Gaussian tails, but provides no additional direct Poisson validation.

Dataset	Bin [MeV]	Raw peak [MeV] / r	RMS a	RMS ΔR	Cond. mass	Cond. Z_g
2015	0.25	51 / 3.139	0.696	0.0000	51	2.083
2015	0.5	51 / 3.039	0.654	0.0023	22	1.978
2015	1	51 / 2.927	0.519	0.0057	51	1.590
2016	0.25	90 / 3.425	4.967	0.0000	42	9.067
2016	0.5	117 / 3.374	4.959	0.0011	43	8.728
2016	1	90 / 3.357	4.933	0.0022	43	8.775
2021	0.625	78 / 2.809	0.807	0.0000	92	1.956
2021	1.25	79 / 3.017	0.803	0.0032	92	2.303
2021	2.5	78 / 3.254	0.785	0.0086	92	1.901

Table 1. Bin width, largest raw root and its mass, full-grid RMS stress offset, RMS change in the correlation matrix relative to baseline, and the selected conditional mass and global normal quantile. Conditional columns test each binning’s specified stress reference. They are not measured particle significances and should not be maximized across rows. Probability estimates and Monte Carlo standard errors are in the CSV.

What a stable correlation matrix would and would not mean

Across the six new matrices, the RMS change in correlation is only 0.0011–0.0086. The matrix describes joint fluctuation structure under its generating background and estimator. Even an accurately computed matrix cannot repair a biased mean or a wrong tail distribution. The particularly large 2016 conditional scores arise because its observed scan is compared with a reference that can predict large fitted deficits. A smoother covariance or a small change in the binning does not turn this discrepancy into a resonance.

The actual rebinned fits and perturbation scans are useful diagnostics of the method. The new coarse fields still need independent complete Poisson or event-bootstrap scans to test their marginal distributions, correlation matrices and maxima. The nine baseline spot checks are numerical checks, not those statistical validation experiments. The 2016 source convergence, development-sample overlap and frozen-state qualifications remain applicable.

The fixed-kernel qualification

Retaining the reviewed kernel coordinates makes this a diagnostic with a fixed kernel policy. A production binning choice still needs independent background closure and injected-signal recovery tests. No preferred binning is selected here.

3. Changing the mass-scan grid is not changing histogram bins

For this test the original histogram and R remain fixed. We draw 100,000 zero-mean, unit-variance Gaussian fields per dataset, then inspect the 1, 2 and 5 MeV subgrids. Every possible phase of each coarse grid is shown; both search boundaries are retained. These are subsets of the frozen 1 MeV grid, not a test of resolution finer than 1 MeV.

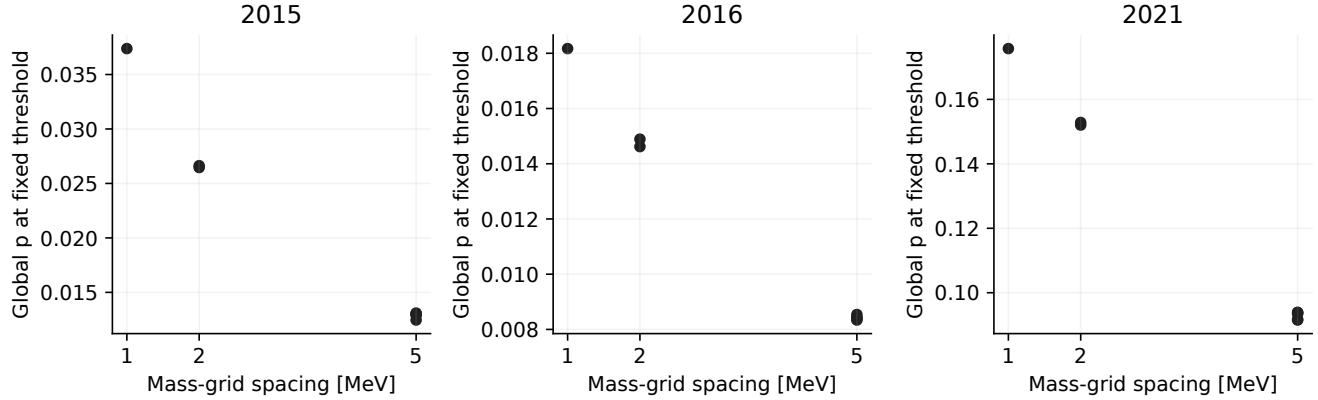


Figure 2. Probability of exceeding the *same original raw-peak threshold* after discarding mass hypotheses. Each dot is a grid phase. Fewer tested points lower this probability, but can also miss the observed peak. All values in this page are conditional on the hypothetical zero-mean Gaussian reference; 2016 does not satisfy its null-mean assumption under the archived stress construction.

Dataset	Step [MeV]	Raw peak r	Global p , grid peak	Global p , fixed peak
2015	1	3.139–3.139	0.0374–0.0374	0.0374–0.0374
2015	2	3.025–3.139	0.0265–0.0375	0.0265–0.0266
2015	5	2.496–3.139	0.0131–0.0897	0.0125–0.0131
2016	1	3.425–3.425	0.0182–0.0182	0.0182–0.0182
2016	2	3.401–3.425	0.0149–0.0158	0.0146–0.0149
2016	5	2.796–3.425	0.0083–0.0659	0.0083–0.0085
2021	1	2.809–2.809	0.1758–0.1758	0.1758–0.1758
2021	2	2.617–2.809	0.1520–0.2455	0.1520–0.1528
2021	5	2.366–2.809	0.0938–0.2887	0.0915–0.0939

Table 2. Ranges across all grid phases. “Grid peak” uses the observed maximum actually retained on that grid. “Fixed peak” keeps the original 1 MeV-grid threshold, isolating loss of trials. These two effects must not be confused. Binomial Monte Carlo intervals and two-sided maximum probabilities are included in the CSV.

The paper discusses loss of upcrossings and peak height on coarse grids [1]. Our explicit subset experiment exhibits both mechanisms. It does not show that a coarse grid is preferable because its fixed-threshold p -value is smaller. A denser scan can find a higher local maximum and also increases the global search opportunity. Resolving the continuum limit would require new fits and response evaluations on a finer mass grid, rather than interpolating the existing observed curve and calling the new peak measured.

4. What if the input count bins are correlated?

For a count covariance V , write $V = LL^T$. A first-order generalization uses independent noise directions given by columns of L :

$$\Delta_{\alpha m} = r_m(B + L_{:\alpha}) - r_m(B), \quad C_r \simeq \Delta^T \Delta = J V J^T, \quad J_{mi} = \frac{\partial r_m}{\partial n_i}.$$

This changes the perturbation basis, not just the plotted error bars. If these excursions would make counts negative or the response is nonlinear, use a controlled derivative step or a valid correlated event generator. Independent complete scans must check the approximation.

As a sensitivity test only, set the correlation of standardized input-bin errors to $Q_{ij} = \rho^{|i-j|}$ for $\rho = 0, 0.1, 0.3$. With $E_{im} = D_{im}/s_m$, the covariance of the original centered score becomes $E^T Q E$. Its diagonal is generally not one. We retain this changed variance when evaluating the original statistic and positive-fit gate; normalizing the diagonal away without adjusting the thresholds would hide the effect.

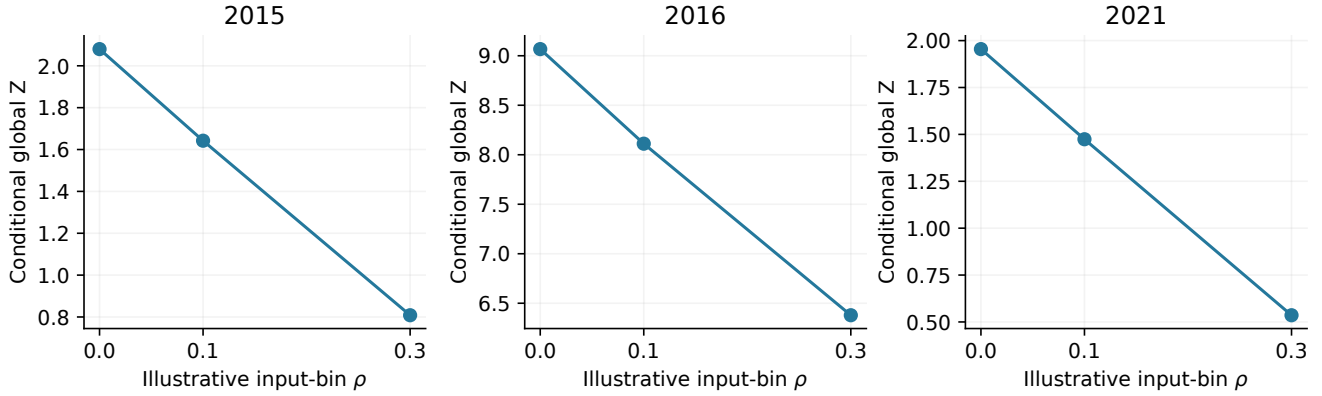


Figure 3. Conditional global normal quantiles at the original selected threshold, with the stress offset and ordering fixed. Each point uses 20,000 importance samples. The correlations are illustrative, not estimates from HPS data. Equal ρ refers to adjacent bins, whose physical widths differ between datasets. In particular, the large 2016 values continue to reflect its problematic stress reference.

Positive neighboring-bin correlations in this example increase score widths and substantially reduce the conditional normal quantiles. This shows why the independence assumption deserves an event-level check; it does not identify a corrected HPS significance. Other covariance structures can have different effects.

The needed input is a count-covariance estimate or generator preserving actual event structure: for example, resample complete events with all their candidates together, or resample suitable acquisition blocks if events are temporally correlated. The aggregate histograms used here do not establish that structure. Correlated GP prediction uncertainty already present in the likelihood should not be added again as raw event-count noise without a generative justification.

5. Which existing p-value field can be used?

The inspected implementation in `hps_gpr/bands.py` defines, at a fixed mass and for an upper limit U ,

$$p_{\text{strong}} = \Pr(U^* \leq U_{\text{obs}}), \quad p_{\text{weak}} = \Pr(U^* \geq U_{\text{obs}}), \quad p_{\text{two}} = \min\{1, 2 \min(p_{\text{strong}}, p_{\text{weak}})\}.$$

These are probabilities for unusually strong or weak *limits*. They are not probabilities for a signal-like excess; both limit construction and background uncertainty affect them. They can define a separate global limit-anomaly test, but that test needs coherent full-scan limit toys and its own joint dependence model. Independent pointwise toy rows cannot supply it.

For the signed discovery root, the usual asymptotic upper tail is $1 - \Phi(r)$ for an excess. A two-sided signed-root test instead uses $2[1 - \Phi(|r|)]$ locally and the maximum of $|r|$ globally. This differs from the two-sided *limit* diagnostic above. A toy-calibrated signed root could be transformed through its null CDF, $z_m = \Phi^{-1}(F_{0,m}(r_m))$, with an explicit treatment of discrete ties. Normal marginals alone do not establish a multivariate Gaussian field: the dependence and maximum distribution must also be validated. A conservative maximum of p-values across background models need not be uniform under any one model; applying an inverse-normal transformation does not fix that.

Dataset	Mass [MeV]	r_{max}	Local p_0	Zero-mean GP	Stress GP	Direct stress
2015	51	3.139	0.00085	0.0374	0.5412	0.539
2016	90	3.425	0.00031	0.0182	1.0000	1.000
2021	78	2.809	0.00249	0.1758	0.7856	0.778

Table 3. Direct use of the observed raw signed peak. The last three columns are full-grid maximum probabilities under, respectively, a hypothetical zero-mean GP, the affine stress GP, and the saved 1,000 complete stress Poisson scans. Both GP columns use 100,000 fields. The very different answers expose the role of the null model. A sampled value of 1 is an estimate at the available resolution. None qualifies the physical background by itself.

Can this locate a more convincing excess?

For one fixed field and search domain, the global tail $\Pr(T^* \geq u)$ decreases with threshold u . The largest local coordinate therefore remains the smallest global p-value when the same ordering is used everywhere. The global correction cannot increase significance relative to its own consistently calibrated local tail. Changing the background, direction, statistic or binning can change the ordering, but adds a model or analysis choice that must be included in validation. The 2016 stress-centered peak at 42 MeV is not more trustworthy merely because it outranks the raw peak.

Practical conclusion. Use the signed discovery statistic, perturb coherent background spectra, propagate the actual count covariance, and validate the complete field under plausible backgrounds. Test injections with the same scan and a common validated global false-positive rate. The present binning comparisons do not establish a new discovery region.

6. 2016 follow-up: would a 32 or 33 MeV support edge help?

We move the GP training lower edge from 30 to 31, 32, 33 and 34 MeV, keeping the upper edge at 210 MeV, the 0.25 MeV bins, all 142 hypotheses from 39 to 180 MeV, and the original reviewed kernel coordinates. The signal windows remain inside every support. Only the stress spectra are fitted. Two shorter fixed length scales and two scaled exposures are additional diagnostic controls; all nine cases were declared before fitting.

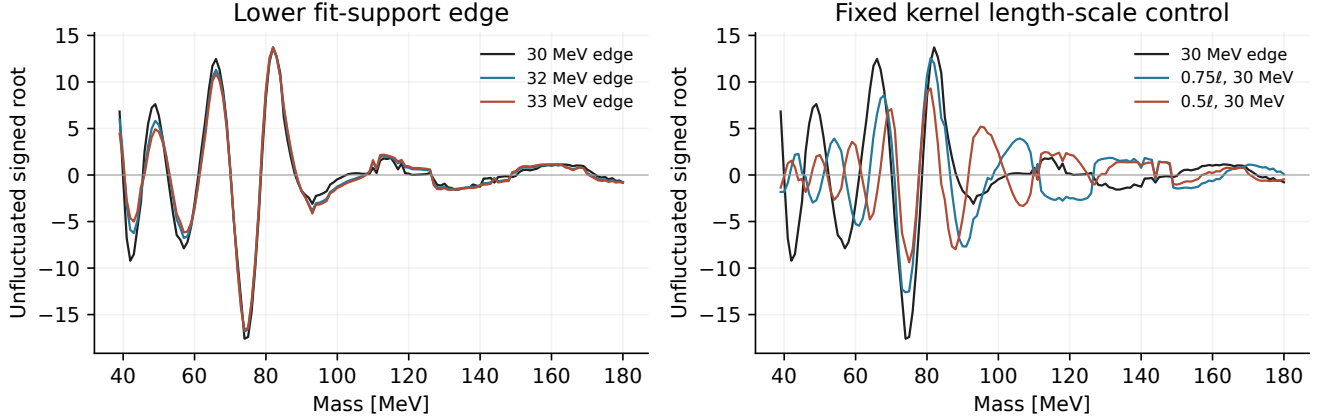


Figure 4. Exact unfluctuated signed roots after changing the lower training edge (left) or multiplying the kernel length scale ℓ by 0.75 or 0.5 (right). Every curve still has substantial bias. The shorter kernels change both the mean prediction and its uncertainty.

Control	All	39–60	61–90	91–180
30 MeV edge	4.967	5.888	9.390	1.033
31 MeV edge	4.816	5.340	9.220	1.136
32 MeV edge	4.670	4.593	9.104	1.278
33 MeV edge	4.580	3.925	9.068	1.386
34 MeV edge	4.553	3.678	9.068	1.412
0.75ℓ , 30 MeV	4.006	2.491	7.580	2.156
0.5ℓ , 30 MeV	3.053	1.796	5.646	1.813
10% exposure, 30 MeV	1.590	2.294	2.822	0.219
1% exposure, 30 MeV	0.546	0.991	0.805	0.122

Table 4. RMS signed stress response over the full scan and three mass ranges in MeV. Exposure controls scale the same generating spectrum and recompute its GP prediction; “10%” is not a new analysis of the actual 2016 development dataset.

The 32 and 33 MeV edges lower the overall RMS from 4.967 to 4.670 and 4.580, improvements of about 6% and 8%. At 33 MeV, the 39–60 MeV RMS improves from 5.888 to 3.925, but the 61–90 MeV RMS remains 9.068 and the 91–180 MeV RMS worsens from 1.033 to 1.386. The worst original deficit, at 74 MeV, changes only from -17.60 to -16.69 .

This is a partial edge effect, not a support-only cure. Halving ℓ reduces the full RMS to 3.053, while worsening some high-mass responses. The earlier 28–34 MeV support study also found no edge passing its full validation gate. Neither the new RMS comparison nor the shorter-kernel control qualifies a replacement analysis or a new global p-value.

7. What makes the 2016 stress response so large?

The baseline has 97.3% of its summed squared response below 90 MeV. The 76 MeV signal window contains 15.77 million expected stress counts: the GP overpredicts its integrated background by only 0.504%, yet the profiled signed root is -14.65 . At 74 MeV the corresponding values are 15.93 million counts, 0.775%, and -17.60 . The root depends on the signal-shaped residual and correlated uncertainty, not just the integrated fractional error. Such small fractional shape errors matter at this occupancy.

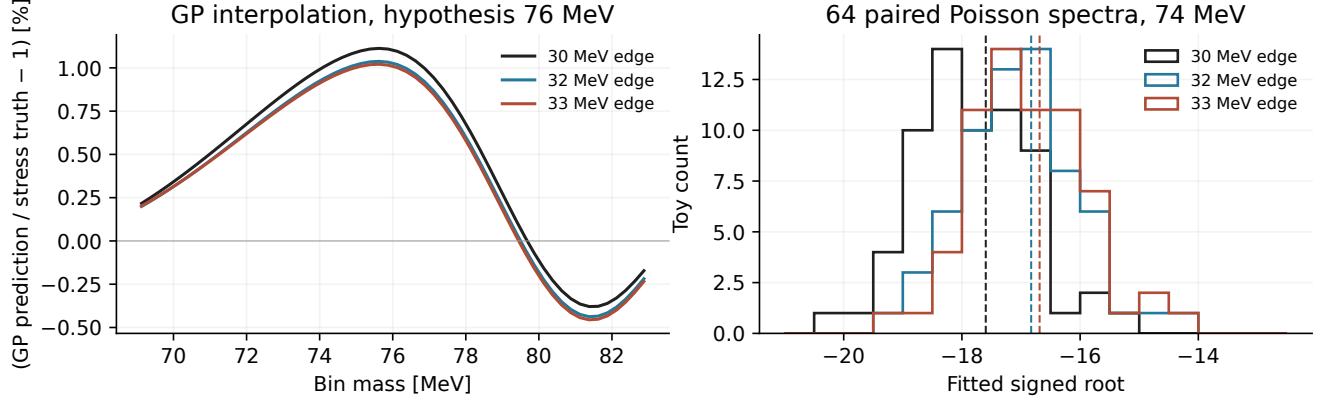


Figure 5. GP prediction relative to the known stress truth across the 76 MeV extraction window (left). Fitted roots at 74 MeV from the same 64 whole Poisson spectra under three support choices (right); dashed lines show the corresponding unfluctuated roots. The large offset survives actual count fluctuations.

Mass	Asimov, 30	Mean, 30	Mean, 32	Mean, 33	Toy SD range
42	-9.21	-9.17	-5.87	-4.58	0.90–0.94
66	12.47	12.55	11.39	10.96	0.87–0.90
74	-17.60	-17.81	-17.04	-16.90	0.95–0.96
76	-14.65	-14.80	-13.91	-13.70	0.89–0.90
83	12.71	12.95	12.86	12.99	0.92–0.92
120	0.15	0.19	0.82	0.97	1.00–1.01

Table 5. Unfluctuated baseline root and toy means for the three lower edges in MeV. Each mean uses 64 paired spectra; standard errors are 0.11–0.13. The final column gives sample standard deviations across support choices. Every toy mean is within 2.11 Monte Carlo standard errors of its own unfluctuated prediction. Paired edge differences and their errors are saved in the CSV.

Reducing the same spectrum to 10% and 1% exposure gives RMS responses 1.590 and 0.546, broadly consistent with statistical amplification of shape mismatch. The GP is refitted at each exposure, so exact square-root scaling is not assumed. The shorter-kernel controls show that interpolation policy also matters. Together these tests explain why a modest fractional residual can yield an outlying 2016 curve; they do not establish the stress truth as the physical background. Its 75–85 MeV blend and broad-component fit qualifications remain, and the earlier blend-removal control worsened the overall response.

These 64 spectra assume independent Poisson input bins and are evaluated at six fixed masses and three supports: 1,152 pointwise fits. They check the mean response locally, not rare tails, complete-scan maxima or signal recovery. The decisive next step is independent background and injection validation of a declared support and kernel policy, rather than selecting the smallest stress RMS.

8. Reproducibility and remaining qualification

The report is a separate derivative; no v5.0.2 note, observed release, production card or reviewed kernel state is modified. The source bundle contains binning scans, new response matrices and coordinate checkpoints, Gaussian sampling ledgers, input hashes, the fixed protocols and scripts. The pointwise rebin pass took 10.1 seconds; the six complete response calculations took 292.3 seconds with a peak resident memory of 243 MiB. The baseline grid and correlated-noise sampling pass took 2.2 seconds and peaked at 1.46 GiB. Each calculation used one process and one linear-algebra thread. No new Poisson toys were generated for these original binning and field calculations. Gaussian Monte Carlo sizes were fixed before evaluation; 20,000 additional importance fields per histogram choice produce Table 1. The support follow-up adds 1,278 deterministic fits and 64 paired Poisson spectra at six masses under three supports. It took 36.4 seconds and 376 MiB peak resident memory, with one process and one linear-algebra thread.

The new response calculations test the Gaussian approximation algebra and numerical fit implementation at both coarse binnings. They do not include fresh complete Poisson validation samples for those binnings. Direct Poisson comparisons in Table 3 use only the saved baseline experiments. The AR(1) noise study is a linear-response sensitivity calculation using the baseline response basis, not a fit of input-bin correlations or a newly correlated event simulation.

Discovery qualification still requires event-level covariance, credible generating backgrounds, complete-scan validation and injection recovery. Any selection across supports, kernels or statistics must be included.

Rebuild commands

```
python3 scripts/rebin_fits.py
python3 scripts/rebinned_response.py
python3 scripts/field_tests.py
python3 scripts/summarize.py
python3 scripts/summarize_support_2016.py
bash scripts/build.sh
python3 scripts/validate.py
```

The exact fit scripts require the original HPS checkout and input ROOT files listed in the manifest. Tables, figures and the PDF can be regenerated from the bundled numerical inputs. The bundle validation checks saved matrices, checkpoints and input hashes. The support fits can be reproduced with `python3 scripts/support_2016.py`.

Sources

[1] V. Ananiev and A. L. Read, *Gaussian Process-based calculation of look-elsewhere trials factor*, JINST 18 (2023) P05041. [Paper](#), [Sections 2 and 2.2](#).

[2] *sigcorr*, software version 5.0.1, pinned commit `a66c08771a81`. [Official source](#). The covariance normalization and Gaussian factors were executable-crosschecked in the preceding v5.0.2 audit. This report retains its distinction between the canonical centered field and the HPS stress-response extension.

[3] A. B. Owen, Y. Maximov and M. Chertkov, *Importance sampling the union of rare events with an application to power systems analysis*, Electronic Journal of Statistics 13 (2019) 231–254. [Paper](#). The conditional-mixture estimator evaluates a fixed finite-grid Gaussian union; its Monte Carlo errors do not include uncertainty in that Gaussian model.

Local sources. Archived global response studies for 2015, 2016 and 2021; the reviewed configuration/state ledger; `bands.py` definitions; and the v5.0.2 *sigcorr* audit. Exact paths and SHA-256 identities are in `inputs/manifest.json` and the validation report.